# Data Collection and Preprocessing Phase

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| Date | 28 January 2026 |
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| Project Title | Uncovering the Hidden Treasures of the Mushroom Kingdom A Classification Analysis |
| Maximum Marks | 6 Marks |

## Data Preprocessing

The images will be preprocessed by resizing, normalizing, augmenting, denoising, adjusting contrast, detecting edges, converting color space, cropping, batch normalizing, and whitening data. These steps will enhance data quality, promote model generalization, and improve convergence during neural network training, ensuring robust and efficient performance across various computer vision tasks.

| **Section** | **Description** |
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| Data Overview | This project uses image datasets of three mushroom species: **Boletus**, **Lactarius**, and **Russula**. The images are collected from various sources including **SmartInternz**, **custom fieldcaptured images**, and platforms like **Kaggle** and **Wikimedia**. This ensures rich visual diversity and robust generalization during training. |
| Resizing | All images are resized to **224×224 pixels** using OpenCV's cv2.resize() function to ensure uniform input dimensions for CNN-based models. |
| Normalization | Pixel values are normalized to the range **[0, 1]** by dividing by 255.0, improving convergence during model training. |

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| Data Augmentation | Using ImageDataGenerator, images are augmented with **random rotation**, **shifts**, **zoom**, **horizontal/vertical flips**, and **fill modes** to avoid overfitting. |
| Denoising | OpenCV's fastNlMeansDenoisingColored() is applied to reduce environmental noise and improve image clarity, especially for field-captured data. |

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| Edge Detection | cv2.Canny() is used for edge detection, helping to emphasize structure, texture, and contour features of different mushroom species. |
| Color Space Conversion | Images are converted from **BGR to HSV** color space using cv2.cvtColor() to better capture color-based patterns across lighting variations. |
| Image Cropping | Manual center cropping is done on some images to focus on the mushroom body and reduce irrelevant background noise, enhancing object recognition. |
| Batch Normalization | BatchNormalization() is applied in the neural network model to stabilize and accelerate the learning process by reducing internal covariate shift. |

**Data Preprocessing Code Screenshots**

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| Loading Data | |

```python
import os
import cv2
import numpy as np
from tensorflow.keras.preprocessing.image import ImageDataGenerator

data_dir = "path/to/mushroom_dataset"
categories = ["Boletus", "Lactarius", "Russula"]

data = []
IMG_SIZE = 224

for category in categories:
    folder = os.path.join(data_dir, category)
    label = categories.index(category)
    for img_name in os.listdir(folder):
        img_path = os.path.join(folder, img_name)
        img = cv2.imread(img_path)
        if img is not None:
            img = cv2.resize(img, (IMG_SIZE, IMG_SIZE))
            data.append([img, label])

```

Resizing

```python
img = cv2.resize(img, (224, 224))
```

Normalization

```python
data = np.array(data, dtype=object)
X = np.array([i[0] for i in data]) / 255.0  # Normalize pixel values
y = np.array([i[1] for i in data])
```

Data Augmentation

```python
train_datagen = ImageDataGenerator(
    rotation_range=30,
    width_shift_range=0.1,
    height_shift_range=0.1,
    zoom_range=0.2,
    horizontal_flip=True,
    vertical_flip=True,
    fill_mode='nearest'
)
```

Denoising

```python
denoised_img = cv2.fastNlMeansDenoisingColored(img, None, 10, 10, 7, 21)
```

Edge Detection

```python
edges = cv2.Canny(img, threshold1=100, threshold2=200)
```

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| Color Space Conversion | `hsv_img = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)` |
| Image Cropping | `cropped_img = img[30:194, 30:194]  # Manual center crop` |
| Batch Normalization | `from tensorflow.keras.layers import BatchNormalization`<br><br>`model.add(BatchNormalization())` |